Rule 2: A ray parallel to the principal axis passes through or appears to be coming from the principal focus (in case of convex mirror) after reflection (Figure 6.5).

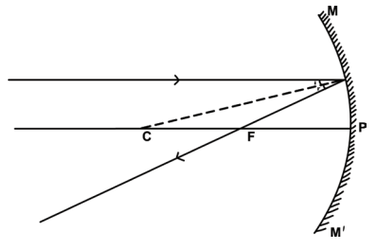


Figure 6.5 Ray parallel to prinicpal axis

Rule 3: A ray passing through the focus gets reflected and travels parallel to the principal axis (Figure 6.6).

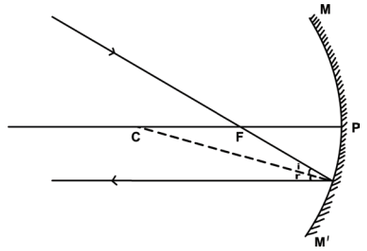


Figure 6.6 Ray travelling through the principal focus

Rule 4: A ray incident at the pole of the mirror gets reflected along a path such that the angle of incidence (APC) is equal to the angle of reflection (BPC) (Figure 6.7).

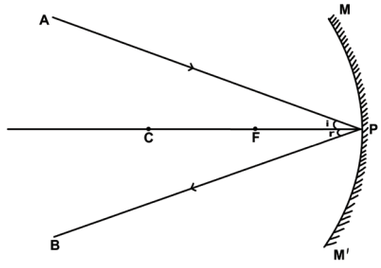


Figure 6.7 Angle of incidence equal to angle of reflection

6.4 Concave Mirror

6.4.1 Image Formation

We shall now find the position, size and nature of image by drawing the ray diagram for a small linear object placed on the principal axis of a concave mirror at different positions.

Case–I: When the object is far away (at infinity), the rays of light reaching the concave mirror are parallel to each other (Figure 6.8a).

Position of the Image: The image is formed at the principal focus F.

Nature of the Image: It is real, inverted and highly diminished in size.

Case–II: When the object is beyond the centre of curvature (Figure 6.8b).

Position of the image: Between the principal focus F and centre of curvature C.

Nature of the image: Real, inverted and smaller than object.

Case – III: When the object is at the centre of curvature (Figure 6.8c).

Position of the image: The image is at the centre of curvature itself.

Nature of the image: It is real, inverted and same size as the object.

Case – IV: When the object is in between the centre of curvature C and principal focus F (Figure 6.8d).

Position of the image: The image is beyond C

Nature of the image: It is real inverted and magnified.

Case – V: When the object is at the principal focus F (Figure 6.8e).

Nature of the image: No image can be captured on the screen nor any virtual image can be seen.

Case – VI: When the object is in between the focus F and the pole P (Figure 6.8f).

Position of the image: The image is behind the mirror.

Nature of the image: It is virtual, erect and magnified.

IX_SCI_EM_Unit-06_PHY.indd 69 30/11/2022 19:25:03